Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 25)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 1 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 1 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 2 . 1 Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

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0 2 . 2 Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

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0 2 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

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0 2 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

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0 3 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 3 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 3 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 3 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer _____ Earths

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0 3 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 3 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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0 4 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

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0 4 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

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0 4 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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0 5 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

0 5 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 5 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

0 6 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

0 6 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

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0 6 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

0 6 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

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0 7 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 7 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

0 7 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

0 7 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

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0 7 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

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0 8 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 8 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

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0 8 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 8 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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0 9 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

0 9 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

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Introduced competitor .....

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Introduced predator .....

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Introduced pathogen .....

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0 9 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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1 0 . 2 State what is meant by a **taxon**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS